



Designation: D7042 – 21a

## Standard Test Method for Dynamic Viscosity and Density of Liquids by Stabinger Viscometer (and the Calculation of Kinematic Viscosity)<sup>1</sup>

This standard is issued under the fixed designation D7042; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon ( $\epsilon$ ) indicates an editorial change since the last revision or reapproval.

### 1. Scope\*

1.1 This test method covers and specifies a procedure for the concurrent measurement of both the dynamic viscosity,  $\eta$ , and the density,  $\rho$ , of liquid petroleum products and crude oils, both transparent and opaque. The kinematic viscosity,  $\nu$ , can be obtained by dividing the dynamic viscosity,  $\eta$ , by the density,  $\rho$ , obtained at the same test temperature.

1.2 The result obtained from this test method is dependent upon the behavior of the sample and is intended for application to liquids for which primarily the shear stress and shear rate are proportional (Newtonian flow behavior).

1.3 The precision has only been determined for those materials, viscosity ranges, density ranges, and temperatures as indicated in Section 15 on Precision and Bias. The test method can be applied to a wider range of materials, viscosity, density, and temperature. For materials not listed in Section 15 on Precision and Bias, the precision and bias may not be applicable.

1.4 The values stated in SI units are to be regarded as standard. No other units of measurement are included in this standard.

1.5 *This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and to determine the applicability of regulatory limitations prior to use.*

1.6 *This international standard was developed in accordance with internationally recognized principles on standardization established in the Decision on Principles for the Development of International Standards, Guides and Recommendations issued by the World Trade Organization Technical Barriers to Trade (TBT) Committee.*

### 2. Referenced Documents

#### 2.1 ASTM Standards:<sup>2</sup>

- D341 Practice for Viscosity-Temperature Equations and Charts for Liquid Petroleum or Hydrocarbon Products
- D396 Specification for Fuel Oils
- D445 Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)
- D975 Specification for Diesel Fuel
- D1655 Specification for Aviation Turbine Fuels
- D2162 Practice for Basic Calibration of Master Viscometers and Viscosity Oil Standards
- D2270 Practice for Calculating Viscosity Index from Kinematic Viscosity at 40 °C and 100 °C
- D4052 Test Method for Density, Relative Density, and API Gravity of Liquids by Digital Density Meter
- D6299 Practice for Applying Statistical Quality Assurance and Control Charting Techniques to Evaluate Analytical Measurement System Performance
- D6300 Practice for Determination of Precision and Bias Data for Use in Test Methods for Petroleum Products, Liquid Fuels, and Lubricants
- D6617 Practice for Laboratory Bias Detection Using Single Test Result from Standard Material
- D6708 Practice for Statistical Assessment and Improvement of Expected Agreement Between Two Test Methods that Purport to Measure the Same Property of a Material
- D6751 Specification for Biodiesel Fuel Blend Stock (B100) for Middle Distillate Fuels
- D7467 Specification for Diesel Fuel Oil, Biodiesel Blend (B6 to B20)
- D7566 Specification for Aviation Turbine Fuel Containing Synthesized Hydrocarbons
- D7915 Practice for Application of Generalized Extreme Studentized Deviate (GESD) Technique to Simultaneously Identify Multiple Outliers in a Data Set

<sup>1</sup> This test method is under the jurisdiction of ASTM Committee D02 on Petroleum Products, Liquid Fuels, and Lubricants and is the direct responsibility of Subcommittee D02.07 on Flow Properties.

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<sup>2</sup> For referenced ASTM standards, visit the ASTM website, [www.astm.org](http://www.astm.org), or contact ASTM Customer Service at [service@astm.org](mailto:service@astm.org). For *Annual Book of ASTM Standards* volume information, refer to the standard's Document Summary page on the ASTM website.

\*A Summary of Changes section appears at the end of this standard

## 2.2 ISO Standards:<sup>3</sup>

**ISO 5725** Accuracy (Trueness and Precision) of Measurement Methods and Results

**ISO 8217** Petroleum products – Fuels (class F)

**ISO/IEC 17025** General Requirements for the Competence of Testing and Calibration Laboratories

## 2.3 Other Documents:<sup>4</sup>

**NIST Technical Note 1297** Guideline for Evaluating and Expressing the Uncertainty of NIST Measurement Results

**DEF STAN 91-091** Turbine Fuel, Kerosene Type, Jet A-1<sup>5</sup>

## 3. Terminology

### 3.1 Definitions:

3.1.1 *density*,  $n$ —mass per unit volume.

3.1.2 *dynamic viscosity* [ $\eta$ ],  $n$ —the ratio between the applied shear stress and rate of shear of a liquid at a given temperature.

3.1.2.1 *Discussion*—It is sometimes called the coefficient of dynamic viscosity or, simply, viscosity. Thus, dynamic viscosity is a measure of the resistance to flow or to deformation of a liquid under external shear forces.

3.1.2.2 *Discussion*—The term dynamic viscosity can also be used in a different context to denote a frequency-dependent quantity in which shear stress and shear rate have a sinusoidal time dependence.

3.1.3 *kinematic viscosity*,  $n$ —the ratio of the dynamic viscosity ( $\eta$ ) to the density ( $\rho$ ) of a liquid at a given temperature.

3.1.3.1 *Discussion*—For gravity flow under a given hydrostatic head, the pressure head of a liquid is proportional to its density ( $\rho$ ). Therefore, the kinematic viscosity ( $\nu$ ) is a measure of the resistance to flow of a liquid under gravity.

3.1.4 *relative density (also called specific gravity (SG))*,  $n$ —the ratio of the density of a material at a stated temperature to the density of a reference material (usually water) at a stated temperature.

### 3.2 Definitions of Terms Specific to This Standard:

3.2.1  $T$  at  $12 \text{ mm}^2/\text{s}$ ,  $^{\circ}\text{C}$ ,  $n$ —the temperature at which the material has a kinematic viscosity of  $12 \text{ mm}^2/\text{s}$ .

3.2.1.1 *Discussion*—Term mostly is associated with jet fuel where  $12 \text{ mm}^2/\text{s}$  is considered a critical viscosity value. The temperature is determined using Practice **D341** interpolation or extrapolation calculations from two kinematic viscosity data points. Other critical viscosity data can similarly be determined for other materials as  $T$  at ( $xx$ )  $\text{mm}^2/\text{s}$ .

## 4. Summary of Test Method

4.1 The test specimen is introduced into the measuring cells, which are at a closely controlled and known temperature. The measuring cells consist of a pair of rotating concentric cylinders and an oscillating U-tube. The dynamic viscosity is determined from the equilibrium rotational speed of the inner

cylinder under the influence of the shear stress of the test specimen and an eddy current brake in conjunction with adjustment data. The density is determined by the oscillation frequency of the U-tube in conjunction with adjustment data. The kinematic viscosity is calculated by dividing the dynamic viscosity by the density.

## 5. Significance and Use

5.1 Many petroleum products, and some non-petroleum materials, are used as lubricants and the correct operation of the equipment depends upon the appropriate viscosity of the liquid being used. In addition, the viscosity of many petroleum fuels is important for the estimation of optimum storage, handling, and operational conditions. Thus, the accurate determination of viscosity is essential to many product specifications.

5.2 Density is a fundamental physical property that can be used in conjunction with other properties to characterize both the light and heavy fractions of petroleum and petroleum products.

5.3 Determination of the density or relative density of petroleum and its products is necessary for the conversion of measured volumes to volumes at the standard temperature of  $15^{\circ}\text{C}$ .

## 6. Apparatus

### 6.1 Stabinger Viscometer<sup>6,7</sup>

6.1.1 *Viscosity Measurement*—The Stabinger viscometer uses a rotational coaxial cylinder measuring system. The outer cylinder (tube) is driven by a motor at a constant and known rotational speed. The low-density inner cylinder (rotor) is held in the axis of rotation by the centrifugal forces of the higher density sample and in its longitudinal position by the magnet and the soft iron ring. Consequently, the system works free of bearing friction as found in rotational viscometers. A permanent magnet in the inner cylinder induces eddy currents in the surrounding copper casing. The rotational speed of the inner cylinder establishes itself as the result of the equilibrium between the driving torque of the viscous forces and the retarding eddy current torque. This rotational speed is measured by an electronic system (Hall effect sensor) by counting the frequency of the rotating magnetic field (see **Fig. 1** and **Fig. 2**, No. 2).

6.1.2 *Density Measurement*—The digital density analyzer uses a U-shaped oscillating sample tube and a system for electronic excitation and frequency counting (see **Fig. 2**, No. 3).

<sup>6</sup> The Stabinger viscometer is covered by a patent. Interested parties are invited to submit information regarding the identification of an alternative to this patented item to the ASTM International headquarters. Your comments will receive careful consideration at a meeting of the responsible technical committee, which you may attend.

<sup>7</sup> The sole source of supply of the apparatus known to the committee at this time is Anton Paar GmbH, Anton-Paar-Str. 20, A-8054 Graz, Austria. If you are aware of alternative suppliers, please provide this information to ASTM International headquarters. Your comments will receive careful consideration at a meeting of the responsible technical committee, which you may attend.

<sup>3</sup> Available from American National Standards Institute (ANSI), 25 W. 43rd St., 4th Floor, New York, NY 10036, <http://www.ansi.org>.

<sup>4</sup> Available from National Institute of Standards and Technology (NIST), 100 Bureau Dr., Stop 1070, Gaithersburg, MD 20899-1070, <http://www.nist.gov>.

<sup>5</sup> Available from IHS, 15 Inverness Way East, Englewood, CO 80112, <http://www.global.ihs.com>.

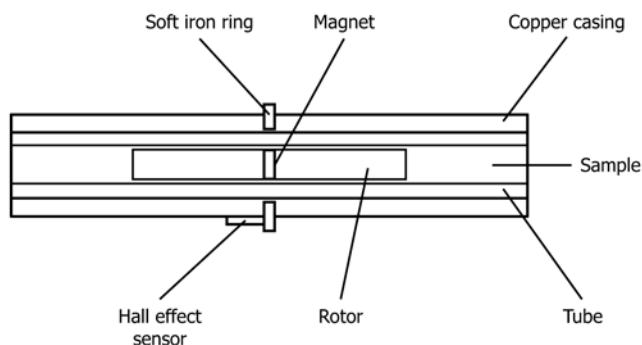


FIG. 1 Viscosity Cell

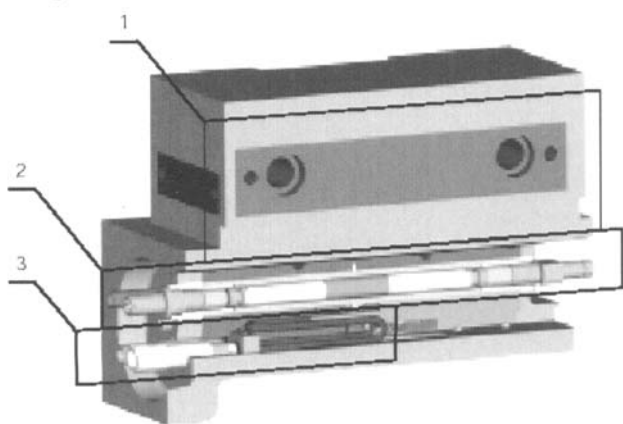


FIG. 2 Cell Block

6.1.3 *Temperature Control*—The copper block surrounds both the viscosity and the density measuring cell in a way that both cells are held at the same temperature. A thermoelectric heating and cooling system (see Fig. 2, No. 1) ensures the temperature stability of the copper block within  $\pm 0.005^\circ\text{C}$  from the set temperature at the position of the viscosity cell over the whole temperature range. The uncertainty ( $k = 2$ ; 95 % confidence level) of the temperature calibration shall be no more than  $\pm 0.03^\circ\text{C}$  over the range from  $15^\circ\text{C}$  to  $100^\circ\text{C}$ . Outside this range the calibration uncertainty shall be no more than  $\pm 0.05^\circ\text{C}$ .

6.1.4 The thermal equilibration time depends on the heat capacity and conductivity of the liquid and on the difference between injection temperature and test temperature. Adequate temperature equilibration of the test specimen is automatically determined when successive viscosity values are constant within  $\pm 0.07\%$  over 1 min and successive density values are constant within  $\pm 0.00003\text{ g/cm}^3$  over 60 s.

NOTE 1—The Stabinger Viscometer, manufactured by Anton Paar GmbH, fulfills the stated requirements when operated in the most precise mode of operation.

6.2 *Syringes*, commercially available, at least 5 mL in volume, with a Luer tip. All construction materials for syringes shall be fully compatible with all sample liquids and cleaning agents, which contact them.

6.3 *Flow-Through or Pressure Adapter*, for use as an alternative means of introducing the test specimen into the mea-

suring cells either by pressure or by suction, provided that sufficient care and control is used to avoid any bubble formation in the test specimen. All construction materials for adaptors shall be fully compatible with all sample liquids and cleaning agents, which contact them.

6.4 *Hot Filling Adapter*, for use with manual syringe filling for the purpose of preventing the precipitation of waxy components dissolved in sample and lowering sample viscosity for easier sample introduction and cleaning routines.

6.5 *Autosampler*, for use in automated injection analyses. The autosampler shall be designed to ensure the integrity of the test specimen prior to and during the analysis and be equipped to transfer a representative portion of test specimen into the measuring cells. The autosampler shall transfer the test specimen from the sample vial to the measuring cells of the apparatus without interfering with the integrity of the test specimen. The autosampler shall be able to mimic the procedure for sample handling as set forth in 11.1 and 11.2. The autosampler may have heating capability as a means to prevent the precipitation of waxy components dissolved in the sample and lower the viscosity of the sample for filling the measuring cells.

6.6 *Screen*, with an aperture of  $75\text{ }\mu\text{m}$ , to remove particles from the sample.

6.7 *Magnet*, strong enough to remove ferromagnetic materials from the sample. Magnetic stirring rods are suitable.

6.8 *Ultrasonic Bath, Unheated (optional)*, with an operating frequency between 25 kHz to 60 kHz and a typical power output of  $\leq 100\text{ W}$ , of suitable dimensions to hold container(s) placed inside of bath, for use in effectively dissipating and removing air or gas bubbles that can be entrained in viscous sample types prior to analysis. It is permissible to use ultrasonic baths with operating frequencies and power outputs outside this range, however it is the responsibility of the laboratory to conduct a data comparison study to confirm that results determined with and without the use of such ultrasonic baths does not materially impact results.

## 7. Reagents and Materials

7.1 *Sample Solvent*, completely miscible with the sample.

7.1.1 For most samples, a volatile petroleum spirit or naphtha is suitable. If the solvent dries up without residues in an applicable time frame, the use of a separate drying solvent is not required.

7.1.2 For residual fuels, a prewash with an aromatic solvent such as toluene or xylene may be necessary to remove asphaltic material.

7.2 *Drying Solvent*, a volatile solvent miscible with the sample solvent (see 7.1).

7.2.1 Highly concentrated ethanol (96 % or higher), n-hexane or n-heptane is suitable.

7.3 *Dry Air or Nitrogen*, for blowing the measuring cells.

7.3.1 If the measuring cell temperature is below or near the dew point temperature of the ambient air, the use of an appropriate desiccator is required.

## 8. Sampling, Test Specimens, and Test Units

### 8.1 General Considerations and Guidelines:

8.1.1 Sampling is defined as all the steps required to obtain an aliquot of the contents of any pipe, tank, or other system, and to place the sample into the laboratory test container. The laboratory test container and sample volume shall be of sufficient capacity to mix the sample and obtain a homogeneous sample for analysis.

8.1.2 For some sample types, such as viscous lube oils that are prone to having entrained air or gas bubbles present in the sample, the use of an ultrasonic bath (see 6.8) without the heater turned on (if so equipped), has been found effective in dissipating bubbles typically within 1 min.

8.1.3 *Particles*—For samples that are likely to contain particles (for example, used oils or crude oils) pass the sample through a 75 µm screen to remove the particles. For the removal of ferromagnetic materials the use of a magnet is appropriate. Waxy samples must be heated to dissolve the wax crystals prior to filtration and a preheated filter shall be used.

8.1.4 *Test Specimen*—A portion or volume of sample obtained from the laboratory sample and delivered to the measuring cells. The test specimen is obtained as follows:

8.1.4.1 Mix the sample, if required, to homogenize. Mixing at room temperature in an open container can result in the loss of volatile material; mixing in closed, pressurized containers, or at sub-ambient temperatures is recommended.

8.1.4.2 Draw the test specimen from a properly mixed laboratory sample using an appropriate syringe. Alternatively, if the proper attachments and connecting tubes are used, the test specimen may be delivered directly to the measuring cells using a flow through or pressure adapter (see 6.3) or autosampler (see 6.5) from the mixing container. For waxy or other samples with a high pour point, before drawing the test specimen, heat the laboratory sample to the desired test temperature, which has to be high enough to dissolve the wax crystals.

### 8.2 Instructions for Residual Fuel Oils:

8.2.1 (**Warning**—Exercise care as vigorous boil-over can occur when opaque liquids which contain high levels of water are heated to high temperatures. Wear appropriate personal protective equipment for handling hot materials.) Place the required number of disposable syringes or samples vials to be used for the batch analysis into a sample preheat apparatus (such as an oven, bath, or heating block) held between 60 °C and 65 °C. If using manual syringe filling, a hot filling adapter must be installed on the apparatus and the injection adapters must also be pre-warmed along with the syringes.

8.2.2 Place the first batch of residual fuel samples to be analyzed for the day in their original containers in a sample preheat apparatus that is between 60 °C and 65 °C for 1 h. Ensure the cap on each container is tightly closed. For samples of a very waxy nature or oils of high kinematic viscosity, it may be necessary to increase the heating temperature above 60 °C to achieve proper mixing. The sample should be sufficiently fluid for ease of stirring and shaking.

8.2.3 Vigorously stir each sample for approximately 20 s with a glass or steel rod of sufficient length to reach the bottom of the container.

8.2.4 Remove the stirring rod and inspect for sludge or wax adhering to the rod. If there is sludge or wax adhering to the rod, continue stirring until the sample is homogeneous.

8.2.5 Recap each container tightly and shake vigorously for 1 min. Then loosen the cap, retighten to finger tight, then back off  $\frac{3}{4}$  to a full turn and place back into the sample preheat apparatus.

8.2.6 Upon completion of 8.2.5 for all samples in the batch, increase the sample pre-heat apparatus temperature to between 100 °C and 105 °C and heat for 30 min.

8.2.7 Remove each container from the sample pre-heat apparatus, close tightly, and shake vigorously for 60 s.

8.2.8 If a heated auto sampler is used, follow instructions in 8.2.8.1 below. If a sample handler is used, follow instructions in 8.2.8.2 below.

8.2.8.1 Ensure the sample vial magazine is held at a temperature between 60 °C and 80 °C. Load each sample into its own preheated sample vial from 8.2.1, insert the vials into the sample vial magazine, and wait for 10 min to 15 min before commencing measurement.

8.2.8.2 Load a preheated sample vial from 8.2.1 with the first sample to be tested in the batch and place the sample vial into the sampler handler that is held between 60 °C and 80 °C. Loosen the cap of the other containers, retighten to finger tight, then back off  $\frac{3}{4}$  to full turn and place the containers back into the sample preheat apparatus that is reset to hold temperature between 60 °C and 80 °C.

8.2.9 Complete the measurements according to 11.1 or 11.3. Analysis of all samples in the batch must be completed within 1 h from completion of 8.2.7.

## 9. Calibration and Verification

9.1 Use only a calibrated apparatus as described in 6.1. The calibration shall be checked periodically using certified reference standards as described in 9.2 and 9.3. The recommended interval for viscosity and density calibration is once a month, for temperature control once a year. For the calibration procedure follow the instructions of the manufacturer of the apparatus.

9.2 *Certified Viscosity and Density Reference Standards*—These are for use as confirmatory checks on the procedure in the laboratory. Certified viscosity and density reference standards shall be certified by a laboratory, which has shown to meet the requirements of ISO/IEC 17025 or a corresponding national standard by independent assessment. Viscosity standards shall be traceable to master viscometer procedures described in Test Method D2162. Density standards shall have a certified uncertainty of the density values of 0.0001 g/cm<sup>3</sup>. The uncertainty of the reference standards shall be stated for each certified value ( $k = 2$ ; 95 % confidence level). See ISO 5725 or NIST Technical Note 1297.

9.3 *Thermometer*—For calibration and adjustment of the temperature control, a digital thermometer with a probe diameter of 6.25 mm and a maximal length of 80 mm shall be used. For smaller probes the use of an adapter is suitable. The uncertainty ( $k = 2$ ; 95 % confidence level) of this thermometer must be no more than  $\pm 0.01$  °C and has to be certified by a laboratory which has shown to meet the requirements of

ISO/IEC 17025 or a corresponding national standard by independent assessment. A suitable thermometer is available from the manufacturer of the apparatus.

**9.4 Acceptable Tolerance**—If the determined values of a calibration check measurement do not agree within the acceptable tolerance band of the certified values, as calculated from **Annex A1**, recheck each step in the procedure, including the special cleaning procedure from **12.2**, to locate the source of error.

**NOTE 2**—Values exceeding the acceptable tolerance are generally attributable to deposits in the measuring cells that are not removed by the routine flushing procedure.

## 10. Adjustment

**10.1** An adjustment has to be carried out when repeated calibration check measurements do not agree with the Acceptable Tolerance Band as stated in **9.4** and the error cannot be located elsewhere. For the adjustment procedure follow the instructions of the manufacturer of the apparatus.

**10.2** For an adjustment, use only certified viscosity and density reference standards that fulfill the requirements as stated in **9.2**. The reference standards have to be within the viscosity, density, and temperature range specified by the manufacturer of the apparatus.

**10.3** After an adjustment procedure a calibration check measurement shall be performed.

## 11. Procedure

**11.1** Standard procedure (rinsing and drying)

**11.1.1** Set the internal temperature control to the desired measuring temperature.

**11.1.2** Set the determinability limits and temperature stability criteria to the values stated in **Table 1** for the specific product.

**11.1.3** Make sure that the measuring cells are clean and dry as described in **12.1**.

**11.1.4** Load a minimum of 5 mL of the test specimen to the syringe; if enough sample is available it is recommended to fill the entire syringe. Pour at least 2 mL of the test specimen into the measuring cells. Leave the syringe in the inlet opening and start the motor and wait for about 10 s, then stop the motor again. Inject an additional 1 mL, leave the syringe in the inlet opening, and start the measurement. Alternatively, the instrument may be configured to perform pre-wetting automatically prior to starting the measurement. Follow manufacture's in-

structions. Wait for the instrument to indicate that the determination is valid and record the values.

**11.1.5** Inject a further 1 mL without taking off the syringe and repeat the measurement.

**11.1.6** If the deviation between two consecutive determinations exceeds the determinability limits as stated in **Table 1** for this product, repeat step **11.1.5** until the deviation is within these limits. Discard all previously determined values and report the values of the last determination as the result.

**11.1.6.1** For products not listed in the precision section, it is the responsibility of the user of this standard to establish reasonable determinability limits by a series of tests.

**11.1.6.2** If the syringe is empty before obtaining a valid determination, rinse and dry the measuring cells as described in **12.1** and repeat step **11.1.4**. If it is not possible to obtain a valid result within an applicable number of repetitions, report the robust mean value and the standard deviation ( $k = 2$ ; 95 % confidence level) together with a remark indicating that the determinability exceeded the limits stated in **15.2.1** for this product.

**11.1.7** Remove the test specimen immediately, rinse and dry the measuring cells as described in **12.1**.

**11.2 Alternative Procedure (Sample Displacement)**—For a series of samples that are mutually soluble (for example, various diesel fuels). It is the responsibility of the user of this standard to determine the applicability of this procedure to each class of samples.

**11.2.1** Set the internal temperature control to the desired measuring temperature.

**11.2.2** Set the determinability limits and temperature stability criteria to the values stated in **Table 1** for the specific product.

**11.2.3** Make sure that the measuring cells are clean and dry as described in **12.1**.

**11.2.4** Load a minimum of 5 mL of the test specimen to the syringe; if enough sample is available it is recommended to use a 10 mL or larger syringe and to fill the entire syringe. Pour slowly at least 3 mL of the test specimen into the measuring cells. A slow flow ensures that the new test specimen displaces the old one rather than merging with it. Leave the syringe in the inlet opening, start the motor and wait for about 10 s, then stop the motor. Inject an additional 1 mL, leave the syringe in the inlet opening, and start the measurement. Alternatively, the instrument may be configured to perform pre-wetting automatically prior to starting the measurement. Follow manufacture's

**TABLE 1 Determinability Limits and Temperature Stability Criteria**

**NOTE 1**—X is the final determination.

|                                                                           | RDV<br>(Determinability,<br>dynamic viscos-<br>ity) | RDD<br>(Determinability,<br>density) | Temperature<br>Stability | Viscosity<br>Stability | Density Stability         | Time | Repeats | Equilibra-<br>tion Time |
|---------------------------------------------------------------------------|-----------------------------------------------------|--------------------------------------|--------------------------|------------------------|---------------------------|------|---------|-------------------------|
| For ALL materials at ALL temperatures<br>unless specifically listed below | 0.10 %<br>0.001 X                                   | 0.0002 g/cm <sup>3</sup>             | ±0.005 °C                | ±0.07 %                | 0.00003 g/cm <sup>3</sup> | 60 s | 3       | 0 s                     |
| Residual Fuel Oils at 50 °C and 100 °C                                    | 0.35 %<br>0.0035 X                                  | 0.0003 g/cm <sup>3</sup>             | ±0.010 °C                | ±0.10 %                | 0.00005 g/cm <sup>3</sup> | 40 s | 3       | 30 s                    |
| Jet fuel at −20 °C and −40 °C, Scanning<br>procedure                      | 0.0011889 X                                         | 0.000113 g/cm <sup>3</sup>           | ±0.005 °C                | ±0.07 %                | 0.0001 g/cm <sup>3</sup>  | 60 s | 0       | 0 s                     |

instructions. Wait for the instrument to indicate that the determination is valid and record the values.

11.2.5 Inject slowly a further 2 mL without taking off the syringe and repeat the measurement.

11.2.6 If the deviation between two consecutive determinations exceeds the determinability limits as stated in **Table 1** for this product, repeat step **11.2.5** until the deviation is within these limits. Discard all previously determined values and report the values of the last determination as the result.

11.2.6.1 For products not listed in the precision section it is the responsibility of the user of this standard to establish reasonable determinability limits by a series of tests.

11.2.6.2 If the syringe is empty before obtaining a valid determination repeat step **11.2.4**. If it is not possible to obtain a valid result within an applicable number of repetitions, report the robust mean value and the standard deviation ( $k = 2$ ; 95 % confidence level) together with a remark indicating that the determinability exceeded the limits stated in **15.1.1** for this product.

11.2.7 For the next sample of this series repeat the steps **11.2.4 – 11.2.6**.

11.2.8 After the last sample of a series perform a cleaning procedure as described in **12.1**.

### 11.3 *Procedure for Use with Autosampler:*

11.3.1 Set the determinability and temperature stability criteria to values in **Table 1** for the corresponding testing parameter(s) and temperature(s) of interest. Ensure that automatic prewetting has been configured. See manufacturer's instructions for details. Temperature control/stability requirements at the test temperatures of interest are provided in **6.1.3**. For all other sample types, see **11.3.2** for guidance.

11.3.2 For products not listed in the precision section, it is the responsibility of the user of this test method to establish reasonable determinability and temperature stability criteria by a series of tests.

11.3.3 Configure the cleaning and drying routines for the autosampler for sufficient cleaning efficiency of the product being tested.

NOTE 3—For specific information on proper configuration, follow the manufacturer's instructions.

11.3.4 Configure the autosampler for a minimum of two consecutive determinations per sample.<sup>8</sup>

11.3.5 Transfer a portion of the sample into the appropriate sample vial. Cap or cover the vials as necessary.

11.3.5.1 Load sample(s) vial onto vial tray or holder and analyze the test specimens.

11.3.5.2 Evaluate the data. Rerun samples which exceed the determinability criteria established for the sample type being analyzed. (See **11.3.1** and **11.3.2**.)

### 11.4 *Procedure for Temperature Scanning:*

11.4.1 The Temperature Scan Procedure is used to study the test material's temperature dependence over a pre-defined temperature range and with pre-defined temperature intervals.

The apparatus reports measured density and dynamic viscosity values as well as calculated kinematic viscosity and the temperature for a pre-defined kinematic viscosity in accordance with Practice **D341**, Standard Viscosity-Temperature Charts for Liquid Petroleum Products. This procedure can be applied with either sample filling procedure.

11.4.2 The temperature range of interest shall be defined as well as the temperature points at which data shall be reported. Ensure that the study material's boiling and freezing points are not exceeded by the defined temperature range. The temperature range and data reporting temperatures shall be entered into the apparatus in either table format or defined as start, stop and interval temperature.

11.4.3 When performing measurements at sub-zero temperatures it is essential that the risk of ice formation during drying routines and sample filling in the measuring cells is eliminated. This can be achieved by using dry air after cell cleaning and selecting a sample filling temperature which is well above 0 °C.

11.4.4 When testing jet fuel in conformance with this test method, the apparatus shall be equipped with an appropriate counter cooling system. The filling temperature shall be 15 °C or above, test temperatures shall be –20 °C, –40 °C with duplicate determinations at –20 °C and –40 °C so that determinability can be verified.

11.4.5 For conformance with stated precision on jet fuel (see **Tables 1-3**) set the test stability criteria in accordance with **Table 1**.

## 12. **Cleaning of the Measuring Cells**

12.1 Fill the measuring cells with at least 3 mL of the sample solvent (see **7.1**). Check if the density cell is filled completely by examining the displayed density value. Start the motor for at least 10 s. For poorly soluble samples, extend this time and repeat the procedure. Then rinse the measuring cells with the drying solvent (see **7.2**) and pass through a stream of dry air or nitrogen until the last trace of solvent is removed. Make sure that the measuring cells are clean and dry by checking if the air density value is lower than 0.0020 g/cm<sup>3</sup>. If the value exceeds this limit repeat the procedure or parts of it.

12.2 *Special Cleaning Procedure*—If necessary, clean the viscosity measuring cell by manually wiping and brushing the measuring rotors using a suitable sample solvent. If necessary clean the density oscillator using suitable cleaning agents. This procedure must be carried out when repeated calibration check measurements do not agree with the acceptable tolerance as stated in **9.4**. Follow the instructions of the manufacturer of the apparatus.

## 13. **Calculation of Results**

13.1 The recorded values are the final results, expressed either as dynamic viscosity in millipascal-seconds or as kinematic viscosity in square millimetre per second and as density in grams per cubic centimetre or kilograms per cubic metre.

13.2 *Viscosity Index*—If results are available from the same sample at 40 °C and at 100 °C the Viscosity Index (VI) can be calculated according to Test Method **D2270** from the kinematic viscosity results at the above temperatures.

<sup>8</sup> The Stabinger Viscometer, manufactured by Anton Paar GmbH, uses the terms, "Repeat Deviation Viscosity, RDV" and "Repeat Deviation Density, RDD" in lieu of "Determinability" in apparatus firmware and documentation.

**TABLE 2 Repeatability (95 %)**

NOTE 1—X is the average of results being compared.

NOTE 2—See Table X3.1 for worked examples for cases where Table 2 contains complex formulas.

| Material                                   | Temperature, °C | Density, g/cm <sup>3</sup> | Dynamic Viscosity, mPa·s | Kinematic Viscosity, mm <sup>2</sup> /s |
|--------------------------------------------|-----------------|----------------------------|--------------------------|-----------------------------------------|
| Base oils                                  | 15              | 0.00046                    | N/A                      | N/A                                     |
| Base oils                                  | 40              | 0.0003                     | 0.00101X or (0.10 %)     | 0.00094X or (0.09 %)                    |
| Base oils                                  | 100             | 0.00033                    | 0.0003516 (X+5)          | 0.0003473 (X+5)                         |
| Formulated oils                            | 40              | 0.0003                     | 0.006279X or (0.63 %)    | 0.006 (X+10.6)                          |
| Formulated oils                            | 100             | 0.0002                     | 0.01209X <sup>0.5</sup>  | 0.002068(X+7.77)                        |
| Diesel fuels                               | 40              | 0.0014 (1.03–X)            | 0.006705X or (0.67 %)    | 0.00782X <sup>0.778</sup>               |
| Jet fuels                                  | –20             | 0.001                      | 0.06477                  | 0.0856                                  |
| Jet fuels, Scanning procedure <sup>A</sup> | –20             | 0.00197X                   | 0.005978X                | 0.007074X                               |
| Jet fuels, scanning procedure              | –40             | 0.00197X                   | 0.00434X                 | 0.005261X                               |
| Biodiesel and biodiesel blends             | 40              | 0.0002                     | 0.004                    | 0.004647                                |
| Residual fuel oils                         | 50              | 0.00098                    | 0.09469X or (9.47 %)     | 0.09093X or (9.09 %)                    |
| Residual fuel oils                         | 100             | 0.00249                    | 0.0517X or (5.17 %)      | 0.05349X or (5.35 %)                    |

<sup>A</sup> This repeatability is currently not referenced in Specification D1655 – 19. Use the precision statement for jet fuel at –20 °C without scanning.

**TABLE 3 Reproducibility (95 %)**

NOTE 1—X is the average of results being compared.

NOTE 2—See Table X3.2 for worked examples for cases where Table 3 contains complex formulas.

| Material                       | Temperature, °C          | Density, g/cm <sup>3</sup> | Dynamic Viscosity, mPa·s    | Kinematic Viscosity, mm <sup>2</sup> /s |
|--------------------------------|--------------------------|----------------------------|-----------------------------|-----------------------------------------|
| Base oils                      | 15                       | 0.00177                    | N/A                         | N/A                                     |
| Base oils                      | 40                       | 0.00147                    | 0.00540X or (0.54 %)        | 0.00584X or (0.58 %)                    |
| Base oils                      | 100                      | 0.00131                    | 0.002563 (X+5)              | 0.002889 (X+5)                          |
| Formulated oils                | 40                       | 0.0013                     | 0.005076X <sup>1.2</sup>    | 0.01087 (X+ 10.6)                       |
| Formulated oils                | 100                      | 0.0012                     | 0.008223X <sup>1.0288</sup> | 0.006346 (X+7.77)                       |
| Diesel fuels                   | 40                       | 0.007 (1.03–X)             | 0.028922X or (2.98 %)       | 0.03374X <sup>0.778</sup>               |
| Jet fuels                      | –20                      | 0.0027                     | 0.1085                      | 0.1485                                  |
| Jet fuels, Scanning procedure  | –20 <sup>A</sup> and –40 | 0.00203X                   | 0.0212X                     | 0.0212X                                 |
| Biodiesel and biodiesel blends | 40                       | 0.0008                     | 0.009595X or (0.96 %)       | 0.009603X or (0.96 %)                   |
| Residual fuel oils             | 50                       | 0.00144                    | 0.1012X or (10.12 %)        | 0.1029X or (10.29 %)                    |
| Residual fuel oils             | 100                      | 0.00499                    | 0.08133X or (8.13 %)        | 0.07186X or (7.18 %)                    |

<sup>A</sup> This reproducibility is currently not referenced in Specification D1655 – 19 for viscosity measurement at –20 °C. Use the precision statement for jet fuel at –20 °C without scanning.

**13.3 Density Extrapolation**—If results are available from the same sample at 40 °C and at 100 °C density values at other temperatures can be calculated by linear extrapolation from the density results at the above temperatures. The limit for this extrapolation temperature is given by the point where changes of state (for example, cloud point or gassing) of the involved sample occur. It is the responsibility of the user of this standard to determine the applicability of this calculation to each class of samples.

## 14. Report

14.1 Report the following information:

14.1.1 The type and identification of the product tested.

14.1.2 The reference to this test method.

14.1.3 Any deviation, by agreement or otherwise, from the procedure specified.

14.1.4 The test results for the dynamic or kinematic viscosity, or both, to four significant figures, optionally also for the density to three decimal places in g/cm<sup>3</sup> or integer in kg/m<sup>3</sup> (example: 0.877 g/cm<sup>3</sup> or 877 kg/m<sup>3</sup>), together with the test temperature.

14.1.5 Date and time of the test.

14.1.6 Name and address of the test laboratory.

## 15. Precision and Bias<sup>9</sup>

15.1 *Comparison of Results:*

15.1.1 *Determinability, (d)*—Quantitative measure of the variability associated with the same operator in a given laboratory, obtaining successive determined values using the same apparatus for a series of operations leading to a single result. It is defined as that difference between two such single determined values as would be exceeded in the long run, in only one case in 20, in the normal and correct operation of the test method.<sup>8</sup> (See Table 1.)

15.2 *Comparison of Results:*

15.2.1 *Repeatability (r)*—The difference between successive results obtained by the same operator in the same laboratory with the same apparatus under constant operating conditions on identical test material would, in the long run, in the normal and correct operation of this test method, exceed the values indicated only in one case in twenty. (See Table 2.)

15.2.2 *Reproducibility (R)*—The difference between two single and independent results obtained by different operators

<sup>9</sup> Supporting data have been filed at ASTM International Headquarters and may be obtained by requesting Research Report RR:D02-1773. Contact ASTM Customer Service at service@astm.org.

working in different laboratories on nominally identical test material would, in the long run, in the normal and correct operation of this test method, exceed the values indicated only in one case in twenty. (See [Table 3](#).)

15.3 *Bias*—No information can be presented on the bias of the procedure in this test method, because no material having an accepted reference value has been tested.

15.4 *Relative Bias*—No bias to Test Methods [D445](#) and [D4052](#) was found to exist for base oils (see [Appendix X1](#) and [Appendix X2](#)).

15.4.1 *Degree of Agreement between Results by Test Method D7042 and Test Method [D445](#)*—Results on the same materials produced by Test Method D7042 and Test Method [D445](#) have been assessed in accordance with procedures outlined in Practice [D6708](#). The findings are:

15.4.2 For formulated oils, RR:D02-1741<sup>10</sup> suggests a bias correction of:

$$Y\text{-Estimate} = 0.996 X \text{ at } 40^\circ\text{C} \quad (1)$$

$$Y\text{-Estimate} = X - 2.3538e - 02 \text{ at } 100^\circ\text{C}$$

where:

Y-Estimate = estimated [D445](#) result using D7042 result.

Bias-corrected kinematic viscosity results from Test Method D7042, as per the bias correction equation that follows, may be considered as practically equivalent to results from Test Method [D445](#), for sample types and property ranges studied. No sample-specific bias, as defined in Practice [D6708](#), was observed after the bias-correction for the materials studied.

$$\text{bias-corrected } X \text{ (predicted } Y) = m X + b; \quad (2)$$

$$m = 0.996;$$

$$b = 0 \text{ at } 40^\circ\text{C}$$

and

$$\text{bias-corrected } X \text{ (predicted } Y) = m X + b; \quad (3)$$

$$m = 1;$$

$$b = -2.3538e - 02 \text{ at } 100^\circ\text{C}$$

where:

X = result obtained by Test Method D7042, and

bias-corrected X = predicted Y result that would have been obtained by Test Method [D445](#) on the same sample.

Differences between bias-corrected results from Test Method D7042 and Test Method [D445](#), for the sample types and property ranges studied, are expected to exceed the following between methods reproducibility (RXY), as defined in Practice [D6708](#), about 5 % of the time. The RXY at 40 mm<sup>2</sup>/s was found to be 0.52156 mm<sup>2</sup>/s at 40 °C and RXY at 10 mm<sup>2</sup>/s was found to be 0.08819 mm<sup>2</sup>/s at 100 °C.

Additional data was requested to ensure that polymer type did not cause differences to the bias between methods. In this pilot study, it was confirmed that application of the previously determined bias correction was acceptable regardless of the polymer type.<sup>11</sup>

15.4.3 For diesel fuels, RR:D02-1742<sup>12</sup> suggests a bias correction of:

$$Y\text{-Estimate} = 0.9872 X - 0.00015 \text{ at } 40^\circ\text{C} \quad (4)$$

where:

Y-Estimate = estimated [D445](#) result using D7042 result.

Bias-corrected kinematic viscosity results from Test Method D7042, as per the bias correction equation that follows, may be considered as practically equivalent to results from Test Method [D445](#), for sample types and property ranges studied. No sample-specific bias, as defined in Practice [D6708](#), was observed after the bias-correction, for the materials studied.

$$\text{bias-corrected } X \text{ (predicted } Y) = m X + b; \quad (5)$$

$$m = 0.9872;$$

$$b = -0.00015$$

where:

X = result obtained by Test Method D7042, and

<sup>11</sup> Supporting data have been filed at ASTM International Headquarters and may be obtained by requesting Research Report RR:D02-1931. Contact ASTM Customer Service at [service@astm.org](mailto:service@astm.org).

<sup>12</sup> Supporting data have been filed at ASTM International Headquarters and may be obtained by requesting Research Report RR:D02-1742. Contact ASTM Customer Service at [service@astm.org](mailto:service@astm.org).

<sup>10</sup> Supporting data have been filed at ASTM International Headquarters and may be obtained by requesting Research Report RR:D02-1741. Contact ASTM Customer Service at [service@astm.org](mailto:service@astm.org).

**TABLE 4 Suggested Relative Bias Correction to Manual [D445](#) Kinematic Viscosity**

| Material            | Temperature °C | Relative Bias Correction, mm <sup>2</sup> /s <sup>A</sup> | Bias-corrected X (predicted Y) = mX+b mm <sup>2</sup> /s |
|---------------------|----------------|-----------------------------------------------------------|----------------------------------------------------------|
| Base Oils           | 40 and 100     | none                                                      | m=1, b=0                                                 |
| Formulated Oils     | 40             | 0.996X                                                    | m=0.996, b=0                                             |
| Formulated Oils     | 100            | X-2.3538e-02                                              | m=1, b=-2.3538e-02                                       |
| Diesel fuels        | 40             | 0.9872X-0.00015                                           | m=0.9872, b=-0.00015                                     |
| Jet fuels           | -20            | 1X-0.013373                                               | m=1, b=-0.013373                                         |
| Biodiesels          | 40             | 0.992X+0.022                                              | m=0.992, b=0.022                                         |
| Residual fuel oils  | 100            | 1X+0.251                                                  | m=1, b=0.251                                             |
| Jet fuels, scanning | -20 and -40    | N/A                                                       | N/A                                                      |

<sup>A</sup> See [15.4](#) for details.

bias-corrected  $X$  = predicted  $Y$  result that would have been obtained by Test Method **D445** on the same sample.

Differences between bias-corrected results from Test Method D7042 and Test Method **D445**, for the sample types and property ranges studied, are expected to exceed the following between methods reproducibility ( $R_{XY}$ ), as defined in Practice **D6708**, about 5 % of the time. The  $R_{XY}$  at 2 mm<sup>2</sup>/s was found to be:

$$\sqrt{(5.5480e - 04 * X^{1.5558e00} + 8.3839e - 04 * Y^{2.6000e00})} \text{ mm}^2/\text{s at } 40^\circ\text{C}$$

15.4.4 For jet fuel at  $-20^\circ\text{C}$ , RR:D02-1750<sup>13</sup> has determined relative bias as follows:

Bias-corrected kinematic viscosity results from Test Method D7042, as per the bias correction equation that follows, may be considered as practically equivalent to results from Test Method **D445**, for sample types and property ranges studied. No sample-specific bias, as defined in Practice **D6708**, was observed after the bias-correction for the materials studied.

$$\text{bias-corrected } X \text{ (predicted } Y) = mX + b; m = 1; b = -0.013373 \quad (6)$$

where:

$X$  = result obtained by Test Method D7042, and

bias-corrected  $X$  = predicted  $Y$  result that would have been obtained by Test Method **D445** on the same sample.

Differences between results from Test Method D7042 and Test Method **D445**, for the sample types and property ranges studied, are expected to exceed the following between methods reproducibility ( $R_{XY}$ ), as defined in Practice **D6708**, about 5 % of the time.  $R_{XY} = 0.162 \text{ mm}^2/\text{s}$  at  $-20^\circ\text{C}$ .

15.4.5 For Biodiesel and Biodiesel blend fuels at  $40^\circ\text{C}$ , RR:D02-1773<sup>14</sup> has determined relative bias as follows:

Degree of Agreement between kinematic viscosity results by Test Method D7042 and Test Method **D445** (manual method)—Results on the same materials produced by Test Method D7042 and Test Method **D445** (manual method) have been assessed in accordance with procedures outlined in Practice **D6708**. The findings are:

**D6708** Outcome: A3

Bias-corrected kinematic viscosity results from Test Method D7042, as per the bias correction equation that follows, may be considered as practically equivalent to kinematic viscosity results from Test Method **D445** (manual method), for sample types and property ranges studied. No sample-specific bias, as defined in Practice **D6708**, was observed after the biascorrection, for the materials studied.

$$\text{bias-corrected } X \text{ (predicted } Y) = 0.992X + 0.022 \quad (7)$$

where:

$X$  = result obtained by Test Method D7042, and

bias-corrected  $X$  = predicted  $Y$  result that would have been obtained by Test Method **D445** (manual method) on the same sample.

Differences between bias-corrected kinematic viscosity results from Test Method D7042 and Test Method **D445** (manual method), for the sample types and property ranges studied, are expected to exceed the following between-methods reproducibility ( $R_{XY}$ ), as defined in Practice **D6708**, about 5 % of the time:

$$\text{Between Methods } R = (0.5 R_X^2 + 0.5 R_Y^2)^{0.5} = (0.0000461 X^2 + 0.000038 Y^{2.8})^{0.5} \quad (8)$$

where:

$R_X$  = reproducibility of D7042 as determined from ILS #895, and

$R_Y$  = reproducibility of **D445** (manual method) as determined from ILS #895.

15.4.6 For residual fuel oils at  $50^\circ\text{C}$ , RR:D02-1837<sup>15</sup> has determined that no bias correction is required and no sample specific biases were identified. For residual fuel oil at  $100^\circ\text{C}$ , RR:D02-1837 has determined a bias correction of  $0.251 \text{ mm}^2/\text{s}$  while no sample specific biases were identified.

$$\text{Bias-corrected } X \text{ (predicted } Y) = 1X + 0.251 \text{ mm}^2/\text{s} \quad (9)$$

Differences between bias-corrected kinematic viscosity results from Test Method D7042 and Test Method **D445** (manual method), for the sample types and property ranges studied, are expected to exceed the following between-methods reproducibility ( $R_{XY}$ ), as defined in Practice **D6708**, about 5 % of the time:

$$\text{Between-methods } R_{XY} = (0.5 R_X^2 + 0.5 R_Y^2)^{0.5} \quad (10)$$

where:

$R_X$  = reproducibility of D7042 as determined from ILS #1003, and

$R_Y$  = reproducibility of **D445** (manual method) as determined from ILS #1003.

15.5 *Interlaboratory Study*—Precision values for base oils were obtained by statistical examination of interlaboratory results obtained by twelve laboratories from ten mineral oils (base oils without additive package) in the dynamic viscosity range from 2.05 mPa·s to 456 mPa·s at  $40^\circ\text{C}$  and from 0.83 mPa·s to 31.6 mPa·s at  $100^\circ\text{C}$  and in the density range from 0.82 g/cm<sup>3</sup> to 0.92 g/cm<sup>3</sup> at  $15^\circ\text{C}$ . Density results at  $15^\circ\text{C}$  were calculated by linear extrapolation from the results obtained at  $40^\circ\text{C}$  and  $100^\circ\text{C}$ . The number of replicates was two, obtained in the shortest possible time interval, typically 10 min, using the standard procedure described in 11.1.

<sup>13</sup> Supporting data have been filed at ASTM International Headquarters and may be obtained by requesting Research Report RR:D02-1750. Contact ASTM Customer Service at service@astm.org.

<sup>14</sup> Supporting data have been filed at ASTM International Headquarters and may be obtained by requesting Research Report RR:D02-1773. Contact ASTM Customer Service at service@astm.org.

<sup>15</sup> Supporting data have been filed at ASTM International Headquarters and may be obtained by requesting Research Report RR:D02-1837. Contact ASTM Customer Service at service@astm.org.

15.5.1 Precision values for base oils were first published in 2003.<sup>16</sup>

NOTE 4—Base oil precision values were determined on instruments using manual sample introduction only.

15.5.2 Precision values for formulated oils (commercially available formulated oils like motor oils, hydraulic oils, and synthetic gear oils) were obtained by statistical examination of interlaboratory results as follows:

15.5.2.1 The dynamic viscosity range from 34 mPa·s to 411 mPa·s at 40 °C from 10 formulated oils by 19 laboratories and from 5.24 mPa·s to 39.2 mPa·s at 100 °C from 10 formulated oils by 18 laboratories.

15.5.2.2 The kinematic viscosity range from 40.164 mm<sup>2</sup>/s to 150.71 mm<sup>2</sup>/s at 40 °C from 9 formulated oils by 19 laboratories and from 6.4682 mm<sup>2</sup>/s to 48.292 mm<sup>2</sup>/s at 100 °C from 10 formulated oils by 18 laboratories.

15.5.2.3 The density range from 0.82989 g/cm<sup>3</sup> to 0.87300 g/cm<sup>3</sup> at 40 °C from 10 formulated oils by 20 laboratories and from 0.79192 g/cm<sup>3</sup> to 0.83512 g/cm<sup>3</sup> at 100 °C from 10 formulated oils by 19 laboratories.

15.5.2.4 Precision values for formulated oils were first published in 2012.

15.5.3 Precision values for diesel fuels were obtained by statistical examination of interlaboratory results obtained by 18 laboratories from eight diesel fuel grades (#1, #2, and #4 diesel fuels according to Specification D975) in the dynamic viscosity range from 1 mPa·s to 6 mPa·s at 40 °C (kinematic viscosity range from 1.6477 mm<sup>2</sup>/s to 7.2315 mm<sup>2</sup>/s at 40 °C) and ten diesel fuel grades (#1, #2, and #4 diesel fuels according to Specification D975) in the density range from 0.76920 g/cm<sup>3</sup> to 0.89174 g/cm<sup>3</sup> at 40 °C.

15.5.3.1 Precision values for diesel fuels were first published in 2012.

15.5.4 Precision values for jet fuel at –20 °C were obtained by statistical examination of interlaboratory results obtained by:

13 laboratories from 12 jet fuel grades (Jet-A, Jet-A1, JP8, SPK, SPK blend, and N2B viscosity standard) in the dynamic viscosity range from 3.1705 mPa·s to 6.7489 mPa·s at –20 °C.

13 laboratories from 12 jet fuel grades (Jet-A, Jet-A1, JP8, SPK, SPK blend and N2B viscosity standard) in the kinematic viscosity range from 3.8087 mm<sup>2</sup>/s to 7.9824 mm<sup>2</sup>/s at –20 °C.

13 laboratories from 12 jet fuel grades (Jet-A, Jet-A1, JP8, SPK, SPK blend and N2B viscosity standard) in the density range from 0.78672 g/cm<sup>3</sup> to 0.85998 g/cm<sup>3</sup> at –20 °C. The degrees of freedom associated with the reproducibility estimate from this round robin study are 18. Since the minimum requirement of 30 (in accordance with Practice D6300) is not met, users are cautioned that the actual reproducibility may be significantly different than these estimates.

15.5.4.1 Precision values for jet fuel at –20 °C were first published in 2012.<sup>13</sup>

15.5.5 Precision values for Biodiesel and Biodiesel blend fuels at 40 °C were obtained by statistical examination of interlaboratory results obtained by:

13 laboratories from 12 biodiesel and biodiesel blend fuels and one synthetic viscosity standard (B100, B6, B10 and B20 from varying feedstock and meeting the specifications of D6751 and D7467) representing a dynamic viscosity range from 1.9818 mPa·s to 4.4196 mPa·s and a kinematic viscosity range from 2.404 mm<sup>2</sup>/s to 5.0767 mm<sup>2</sup>/s.

17 laboratories from 12 biodiesel and biodiesel blend fuels and one synthetic viscosity standard (B100, B6, B10 and B20 from varying feedstock and meeting the specifications of D6751 and D7467) representing a density range from 0.8237 g/cm<sup>3</sup> to 0.8704 g/cm<sup>3</sup>.

NOTE 5—The degrees of freedom associated with the reproducibility estimate for density is 19. Since the minimum requirement of 30 (in accordance with Practice D6300) is not met, users are cautioned that the actual reproducibility may be significantly different than these estimates.

15.5.5.1 Precision values for biodiesel and biodiesel blend fuels at 40 °C were first published in 2014.

15.5.6 Precision values for residual fuel oils at 50 °C and 100 °C were obtained by statistical examination of interlaboratory results obtained by 10 laboratories from 10 residual fuel samples conforming to D396 Grades No. 5 or 6 and/or ISO 8217 RMG and RMK representing a dynamic viscosity range from 25.31 mPa·s to 2212 mPa·s at 50 °C and 5.493 mPa·s to 109.1 mPa·s at 100 °C.

15.5.6.1 The kinematic viscosity range spanned 27.6 mm<sup>2</sup>/s to 2230 mm<sup>2</sup>/s at 50 °C and 6.202 mm<sup>2</sup>/s to 108.1 mm<sup>2</sup>/s at 100 °C.

15.5.6.2 The density range spanned 0.09173 g/cm<sup>3</sup> to 0.9937 g/cm<sup>3</sup> at °C and 0.8831 g/cm<sup>3</sup> to 0.9611 g/cm<sup>3</sup> at 100 °C.

15.5.7 Precision values for jet fuel at –20 °C, –40 °C using the temperature scanning procedure were obtained by statistical examination of interlaboratory results obtained by 10 laboratories from 15 jet fuel samples conforming to D1655, D7566, or DEF STAN 91-091. The samples represented a density range from 0.78313 g/cm<sup>3</sup> to 0.85331 g/cm<sup>3</sup>, a dynamic viscosity range from 2.75465 mPa·s to 5.79955 mPa·s and a kinematic viscosity range from 3.3837 mm<sup>2</sup>/s to 6.7841 mm<sup>2</sup>/s at –20 °C.

The samples represented a density range from 0.79744 g/cm<sup>3</sup> to 0.86796 g/cm<sup>3</sup>, a dynamic viscosity range from 5.2594 mPa·s to 14.002 mPa·s and a kinematic viscosity range from 6.35075 mm<sup>2</sup>/s to 16.1705 mm<sup>2</sup>/s at –40 °C.

15.5.7.1 The interlaboratory studies were performed according to the Practices D6300 and D7915.

15.5.7.2 Precision values for jet fuel at –20 °C and –40 °C using the temperature scanning method were first published in 2019.<sup>17</sup>

15.5.8 The interlaboratory studies were performed according to Practices D6300 and D6708. For the determination of the relative bias the samples were also tested by kinematic

<sup>16</sup> Supporting data have been filed at ASTM International Headquarters and may be obtained by requesting Research Report RR:D02-1555. Contact ASTM Customer Service at service@astm.org.

<sup>17</sup> Supporting data have been filed at ASTM International Headquarters and may be obtained by requesting Research Report RR:D02-1920. Contact ASTM Customer Service at service@astm.org.

viscometers according to Test Method **D445** and for base oils by digital density meters according to Test Method **D4052** in the same laboratories.

## 16. Keywords

16.1 critical viscosity; density; dynamic viscosity; kinematic viscosity; Stabinger viscometer; temperature scanning; viscosity; viscosity index

## ANNEX

### (Mandatory Information)

#### A1. CALCULATION OF ACCEPTABLE TOLERANCE ZONE<sup>18</sup> (BAND) TO DETERMINE CONFORMANCE WITH A CERTIFIED REFERENCE MATERIAL

A1.1 Determine the standard deviation for site uncertainty,<sup>19</sup>  $\sigma_{\text{site}}$ , from a laboratory quality control program.

A1.1.1 If the standard deviation for site uncertainty,  $\sigma_{\text{site}}$ , is not known, use the value 0.30 % for viscosity and 0.0005 g/cm<sup>3</sup> for density.

A1.2 Determine the combined extended uncertainty, CEU, of the accepted reference value (ARV) of the certified reference material (CRM) from the supplier's label or included documentation. If the density uncertainty is not known use the value 0.0005 g/cm<sup>3</sup> for the CEU.

A1.3 Calculate the standard error of the accepted reference value,  $SE_{\text{ARV}}$ , by dividing the CEU by the coverage factor,  $k$ , listed on the supplier's label or included documentation.

A1.3.1 If the coverage factor,  $k$ , is not known, use the value 2.

A1.4 Construct the acceptable tolerance zone:

$$TZ = \pm 1.44 \sqrt{\sigma_{\text{site}}^2 + SE_{\text{ARV}}^2} \quad (\text{A1.1})$$

A1.4.1 Worked out example for viscosity tolerance band:

$$\sigma_{\text{site}} = 0.30 \% \quad (\text{A1.2})$$

$$CEU_{\text{ARV}} = 0.35 \%$$

$$TZ = \pm 1.44 \sqrt{0.30^2 + 0.175^2} = \pm 1.44 \sqrt{0.09 + 0.03} = \pm 0.50 \%$$

A1.4.2 Worked out example for density tolerance band:

$$\sigma_{\text{site}} = 0.0005 \text{ g/cm}^3 \quad (\text{A1.3})$$

$$CEU_{\text{ARV}} = 0.0005 \text{ g/cm}^3$$

$$TZ = \pm 1.44 \sqrt{0.0005^2 + 0.00025^2} = \pm 1.44 \times 0.00056 = \pm 0.0008 \text{ g/cm}^3$$

<sup>18</sup> These calculations are based on Test Method **D6617**.

<sup>19</sup> See Test Method **D6299**.

## APPENDIXES

### (Nonmandatory Information)

#### X1. KINEMATIC VISCOSITY BIAS AT 100 °C COMPARED TO TEST METHOD **D445**

X1.1 The bias calculation in accordance with Test Method **D6708** applied to the results of the interlaboratory study on base oils (see **15.5**) suggested a bias correction of ( $Y = 0.998 X$ ) at 100 °C. This corresponds to a deviation of 0.2 % of

the measured value. A comparison to the precision data from this test method and from the reference test method shows that the suggested bias correction is less than even the repeatability of both test methods over the whole viscosity range tested in

**TABLE X1.1 Kinematic Viscosity Precision versus Bias (Base Oils at 100 °C)**

NOTE 1—X is the average of results being compared.

|                                      | Test Method <b>D445</b> | Test Method <b>D445</b> at this study                  | Test Method <b>D7042</b> |
|--------------------------------------|-------------------------|--------------------------------------------------------|--------------------------|
| Reproducibility, $R$ (95 %)          | 0.0065 X (0.65 %)       | 0.008916 X (0.89 %)                                    | 0.002889 (X+5)           |
| Repeatability, $r$ (95 %)            | 0.0011 X (0.11 %)       | 0.002299 X (0.23 %)                                    | 0.000347 (X+5)           |
| Bias against Test Method <b>D445</b> |                         |                                                        | 0.998 X (0.2 %)          |
| Reference standard oil uncertainty   |                         | 0.30 % to 0.32 % for the viscosity range of this study |                          |

this interlaboratory study and also less than the uncertainty of certified viscosity reference standards (see [Table X1.1](#)). Therefore the bias can be considered insignificant.

X1.2 The kinematic viscosity bias at 40 °C was found to be 1.000 X, corresponding to 0.0 %.

## X2. DENSITY BIAS AT 40 °C COMPARED TO TEST METHOD [D4052](#)

X2.1 The bias calculation in accordance with Practice [D6708](#) applied to the results of the interlaboratory study (see [15.5](#)) suggested a bias correction of ( $Y = 1.0053 X - 0.0046 \text{ g/cm}^3$ ) at 40 °C. This corresponds to a maximum deviation of  $0.0003 \text{ g/cm}^3$  over the density range tested in this interlaboratory study. A comparison to the precision data from this test method and from the reference test method shows that the suggested bias correction is less than even the repeatability

of this test method over the whole density range tested in this interlaboratory study and also less than the uncertainty of certified viscosity reference standards (see [Table X2.1](#)). Therefore the bias can be considered insignificant.

X2.2 The density bias at 15 °C was found to be  $0.0000 \text{ g/cm}^3$ .

**TABLE X2.1 Density Precision versus Bias (Base Oils at 40 °C)**

NOTE 1—X is the average of results being compared.

|                                                | Test Method <a href="#">D4052</a> | Test Method <a href="#">D4052</a> at this study | Test Method <a href="#">D7042</a>   |
|------------------------------------------------|-----------------------------------|-------------------------------------------------|-------------------------------------|
| Reproducibility, <i>R</i> (95 %)               | 0.0005 g/cm <sup>3</sup>          | 0.00044 g/cm <sup>3</sup>                       | 0.0015 g/cm <sup>3</sup>            |
| Repeatability, <i>r</i> (95 %)                 | 0.0001 g/cm <sup>3</sup>          | 0.00010 g/cm <sup>3</sup>                       | 0.0003 g/cm <sup>3</sup>            |
| Bias against Test Method <a href="#">D4052</a> |                                   |                                                 | 1.0053 X – 0.0046 g/cm <sup>3</sup> |
| Reference standard oil uncertainty             |                                   | Typically 0.0005 g/cm <sup>3</sup>              |                                     |

### X3. WORKED EXAMPLES

X3.1 See [Table X3.1](#) and [Table X3.2](#).

**TABLE X3.1 Worked Examples for Repeatability Values at Levels of Viscosity in Cases where [Table 2](#) on Repeatability Contains Complex Formulas**

| Viscosity Level                                | 5      | 10     | 25     | 50     | 100    | 2000   |
|------------------------------------------------|--------|--------|--------|--------|--------|--------|
| Base oils, dynamic viscosity at 100 °C         | 0.07 % | 0.05 % | 0.04 % | 0.04 % | 0.04 % | 0.04 % |
| Base oils, kinematic viscosity at 100 °C       | 0.07 % | 0.05 % | 0.04 % | 0.04 % | 0.04 % | 0.03 % |
| Formulated oils, dynamic viscosity at 100 °C   | 0.54 % | 0.38 % | 0.24 % | 0.17 % | 0.12 % | 0.03 % |
| Formulated oils, kinematic viscosity at 40 °C  | 1.87 % | 1.24 % | 0.85 % | 0.73 % | 0.66 % | 0.60 % |
| Formulated oils, kinematic viscosity at 100 °C | 0.53 % | 0.37 % | 0.27 % | 0.24 % | 0.22 % | 0.21 % |
| Viscosity level                                | 1      | 2      | 3      | 4      | —      | —      |
| Diesel fuel, kinematic viscosity at 40 °C      | 0.78 % | 0.67 % | 0.61 % | 0.57 % | ...    | ...    |

**TABLE X3.2 Worked Examples for Reproducibility Values at Levels of Viscosity in Cases where [Table 3](#) on Reproducibility Contains Complex Formulas**

| Viscosity Level                                | 5      | 10     | 25     | 50     | 100    | 2000   |
|------------------------------------------------|--------|--------|--------|--------|--------|--------|
| Base oils, dynamic viscosity at 100 °C         | 0.51 % | 0.38 % | 0.31 % | 0.28 % | 0.27 % | 0.26 % |
| Base oils, kinematic viscosity at 100 °C       | 0.58 % | 0.43 % | 0.35 % | 0.32 % | 0.30 % | 0.29 % |
| Formulated oils, dynamic viscosity at 40 °C    | 0.70 % | 0.80 % | 0.97 % | 1.11 % | 1.28 % | 2.32 % |
| Formulated oils, dynamic viscosity at 100 °C   | 0.71 % | 0.70 % | 0.70 % | 0.69 % | 0.69 % | 0.69 % |
| Formulated oils, kinematic viscosity at 40 °C  | 3.39 % | 2.24 % | 1.55 % | 1.32 % | 1.20 % | 1.09 % |
| Formulated oils, kinematic viscosity at 100 °C | 1.62 % | 1.13 % | 0.83 % | 0.73 % | 0.68 % | 0.64 % |
| Viscosity level                                | 1      | 2      | 3      | 4      | —      | —      |
| Diesel fuel, kinematic viscosity at 40 °C      | 3.37 % | 2.89 % | 2.64 % | 2.48 % | ...    | ...    |

### SUMMARY OF CHANGES

Subcommittee D02.07 has identified the location of selected changes to this standard since the last issue (D7042 – 21) that may impact the use of this standard. (Approved Nov. 1, 2021.)

(1) Revised subsections [6.7](#), [7.2.1](#), [8.1.3](#), [8.2.4](#), [11.1.4](#), [11.2.4](#), [11.3.1](#), and [14.1.4](#).

Subcommittee D02.07 has identified the location of selected changes to this standard since the last issue (D7042 – 20) that may impact the use of this standard. (Approved Jan. 1, 2021.)

(1) Revised definition for density in subsection [3.1.1](#).

Subcommittee D02.07 has identified the location of selected changes to this standard since the last issue (D7042 – 19<sup>e1</sup>) that may impact the use of this standard. (Approved June 1, 2020.)

(1) Revised definitions for density, dynamic viscosity, and kinematic viscosity in Section [3](#).

(2) Revised subsection [15.4.2](#) to add footnote 11 and RR:D02-1931.

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